Employees

EmployeeID	FirstName	LastName	Department	Salary	City	Age
1	John	Doe	IT	70000	New York	35
2	Alice	Smith	HR	60000	Los Angeles	40
3	Bob	Johnson	Finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	Finance	80000	New York	50

Projects

Project_ID	EmployeeID	ProjectName	Budget	Start Date	Status
101	1	AI Development	100000	2024-01-10	Completed
102	2	Employee Training	50000	2024-02-15	Ongoing
103	1	Cybersecurity Audit	75000	2024-03-05	Pending
104	3	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	80000	2024-06-10	Pending

① Question: Retrieve all employees and their assigned projects using an INNER JOIN.

. Return: EmployeeID, FirstName, LastName, Department, Salary, ProjectID, ProjectName, Budget, Status.

```
SELECT EmployeeID, FirstName, LastName, Department, Salary, ProjectID,
    ProjectName, Budget, Status
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk management	80000	Pending

② Retrieve all employees and their assigned projects, including employees who have no projects using a LEFT JOIN.

```
SELECT EmployeeID, FirstName, LastName, Department, Salary, ProjectID,
       ProjectName, Budget, Status
FROM Employees As A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Management	80000	Pending

③ Retrieve all projects and their assigned employees, including projects that have no employees using a RIGHT JOIN.

```
SELECT ProjectID, ProjectName, Budget, Status, EmployeeID, FirstName,
       LastName, Department, Salary
FROM Employees AS A
RIGHT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

ProjectID	ProjectName	Budget	Status	EmployeeID	FirstName	LastName	Department	Salary
101	AI Development	100000	Completed	1	John	Doe	IT	70000
102	Employee Training	50000	Ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity Audit	75000	Pending	1	John	Doe	IT	70000
104	Financial Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Expansion	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Management	80000	Pending	6	Michael	Clark	Finance	80000

④ Retrieve all employees and projects, including those without a match in either table using a FULL OUTER JOIN

```
SELECT EmployeeID, FirstName, LastName, Department, Salary, ProjectID,
       ProjectName, Budget, Status
FROM Employees AS A
~~OH Projects AS B~~ FULL OUTER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	101	AI Development	100000	Completed
1	John	Doe	IT	70000	103	Cybersecurity Audit	75000	Pending
2	Alice	Smith	HR	60000	102	Employee Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analysis	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	Market Expansion	65000	Complete
6	Michael	Clark	Finance	80000	106	Risk Management	50000	Pending
4	Daviel	Brown	IT	72000	NULL	NULL	NULL	NULL

3. FILTERING

① Retrieve employees who earn more than 70000

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Salary > 70000;
```

EmployeeID	FirstName	lastName	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

⑧ Retrieve employees working in either IT or Finance departments

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
WHERE Department = 'IT'
  OR Department = 'Finance';
```

EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

⑨ Retrieve projects that are not yet completed

```
SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Status != 'completed';
```

ProjectID	ProjectName	Budget	Status
102	Employee Training	50000	Ongoing
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

⑩ Retrieve projects that have a budget greater than 70000 and are not completed

```
SELECT ProjectID, ProjectName, Budget, Status
FROM Projects
WHERE Budget > 70000
AND Status != 'completed';
```

ProjectID	ProjectName	Budget	Status
103	Cybersecurity Audit	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

(11) Retrieve employees from New York OR Toronto, ordered by salary in descending order

```
SELECT EmployeeID, FirstName, LastName, Department, Salary, City
FROM Employees
WHERE City IN ('New York', 'Toronto')
ORDER BY Salary DESC;
```

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

(12) Retrieve the top 3 highest paid employees

```
SELECT EmployeeID, FirstName, LastName, Department, Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;
```

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

4 GROUP BY & HAVING

(13) Find the total salary per department, sorted in descending order

```
SELECT Department, sum(salary) As TotalSalary
FROM Employees
GROUP BY Department
ORDER BY DESC;
```

Department	TotalSalary
Finance	155000
IT	142000
Sales	65000
HR	60000

(14) Find the average salary per city, but only include cities where the average salary is greater than 65 000

```
SELECT City, Avg (Salary) as AverageSalary
FROM Employees
GROUP BY City
HAVING Avg (Salary) > 65 000;
```

City	AverageSalary
New York	75000
Toronto	75000
London	72000

(15) Count the number of employees per department, including only departments with more than 1 employee

```
SELECT Department, Count (Employee ID) As Employee Count
FROM Employees
GROUP BY Department
HAVING Count (EmployeeID) >1;
```

Department	Employee Count
IT	2
Finance	2

⑩ Retrieve the number of projects per status, but only include statuses with at least 2 projects

```
SELECT status, Count (ProjectID) AS Project Count
FROM Projects
GROUP BY status
HAVING Count (Project ID) >= 2;
```

Status	Project Count
Completed	2
Ongoing	2
Pending	2